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SPHEREx Blink

A blink viewer for the public SPHEREx images

Release v1.0.0, technical documentation

Team **Observationally Speaking**

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Source code `https://github.com/yourastrophysicist/SPHEREx-site`
Project page `https://www.yourastro.space/SPHEREx/`

Abstract

SPHEREx Blink is a public web tool that shows the same small patch of sky in every SPHEREx visit, one after the other, like a flipbook. The user gives a position on the sky, and optionally a proper motion. The tool asks the NASA/IPAC Infrared Science Archive (IRSA) which SPHEREx exposures cover that position, downloads small cutouts, subtracts the zodiacal light, flags bad pixels and resamples every exposure onto one fixed north-up grid. On that grid the field stars stand still, so anything that moves or changes brightness stands out to the eye. The design follows WiseView, the blink tool of Backyard Worlds: Planet 9. Every setting lives in the page URL, so one link is one view and can be shared. This document describes release v1.0.0: what the tool does, how the data path works, how to run it, how it was tested on the five targets named by the challenge, and what it cannot do.

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1 The challenge and our answer

The challenge asked for a public-facing web tool. The user enters a right ascension and a declination, and the tool plays the SPHEREx images of that patch as a short video, so that moving objects are easy to spot. It named five test targets: Barnard's Star, Epsilon Indi B, WISE 0855−0714, RW Aurigae and R Draconis. It pointed to WiseView as a model.

SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) was launched on 12 March 2025. It maps the whole sky every six months with six detectors between 0.75 μm and 5.0 μm. Each detector sits behind a linear variable filter, so the wavelength changes across the array, and one star seen in two exposures is in general seen at two slightly different wavelengths. The pixels are large, about 6.15″. This makes SPHEREx a good survey for slow, bright, infrared movers such as nearby stars and brown dwarfs, and a hard one for faint fast objects.

Our answer has three parts.

1. A **data path** in Python, one function call from coordinates to a list of aligned frames (Section 2).
2. A small **server** that wraps that call in one JSON endpoint and serves the page (Section 3).
3. A **web page** that blinks the frames with WiseView-style controls and keeps every setting in the URL (Section 4).

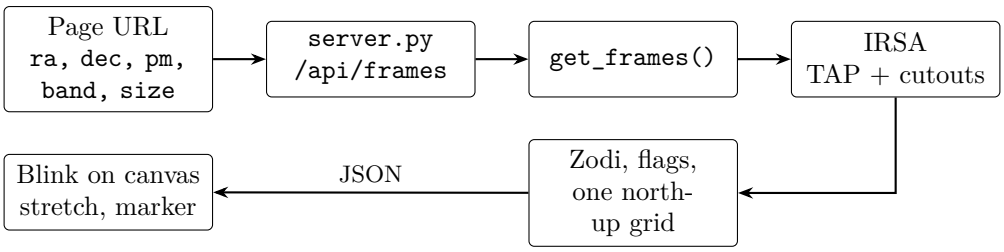


Figure 1: The path from a link to the blink.

2 The data path

All of this is in `tool/spherex_frames.py`. The public entry point is

```
frames, grid = get_frames(ra, dec, band="D1", fov=0.03, epoch=2016.0,
                          pmra=0.0, pmdec=0.0, name="target", mask=True)
```

It returns the common grid as an `astropy` WCS and a list of frames, oldest first. Each frame holds the image on the grid, the variance, the date (MJD and decimal year), the wavelength at the target, the index of the six-month pass it belongs to, the predicted pixel of the target at that date, and the archive URI of the exposure.

2.1 Which exposures

The archive is queried through IRSA's TAP service with ADQL. A row is kept when the footprint polygon of the exposure contains the target and the detector matches:

```
SELECT a.uri, p.time_bounds_lower
FROM spherex.artifact a JOIN spherex.plane p ON a.planeid = p.planeid
WHERE 1 = CONTAINS(POINT('ICRS', ra, dec), p.poly)
  AND p.energy_bandpassname = 'SPHEREx-D1'
ORDER BY p.time_bounds_lower
```

The position used in the query is the target's position at 2026.0. The answer is cached on disk for 24 hours, because the query alone takes about 20 seconds and new SPHEREx data reach the archive about once a week.

2.2 Moving the target to each date

Positions from Gaia DR3 are given at epoch J2016.0. A star with a large proper motion is many SPHEREx pixels away from that position by the time SPHEREx looks at it, so the position is moved to the date of each exposure. Over a few years a linear motion is enough:

$$\alpha(t) = \alpha_0 + \frac{\mu_{\alpha*}\,(t - t_0)}{\cos\delta_0}, \qquad \delta(t) = \delta_0 + \mu_\delta\,(t - t_0), \tag{1}$$

with (α_0, δ_0) at epoch t_0, $\mu_{\alpha*} = \mu_\alpha \cos\delta$ and μ_δ in $\mathrm{mas\,yr^{-1}}$, converted to degrees. Parallax is not applied. For Barnard's Star the parallax is $0.55\,''$, about a tenth of a pixel, and it shows up as a small wobble on top of the straight walk.

2.3 Cutouts

For each exposure a cutout centred on the moved position is asked from IRSA's image service. The cutout side is the field of view plus the distance the target travels over the whole time span plus two pixels:

$$s = \mathrm{fov} + \Delta t \sqrt{\mu_{\alpha*}^2 + \mu_\delta^2} + 2\,p, \qquad p = 6.15'', \tag{2}$$

so that the fixed grid is covered at every date even though the cutout centre moves. Downloads run in four parallel threads. The archive sometimes answers with a 503 or sends a file cut short, so each download is checked for the five extensions the tool needs (`IMAGE`, `FLAGS`, `VARIANCE`, `ZODI`, `WCS-WAVE`) and retried with an exponential back-off up to eight times. One exposure that still fails is skipped and logged instead of stopping the whole view.

2.4 Cleaning each exposure

- **Zodiacal light.** The `ZODI` extension, the pipeline's model of the zodiacal light, is subtracted from `IMAGE`. Without this the sky level changes from visit to visit and the blink flickers.
- **Bad pixels.** A pixel is flagged when any of the flag bits 0, 1, 6, 9, 10, 11, 15, 17 or 19 is set, the same set used in our prep script `coadd_spherex.py`. With `mask=True` flagged pixels become NaN. The viewer asks with `mask=0` by default, because the saturated core of a bright star is flagged and a blink is easier to read when the star is still drawn.
- **Wavelength at the target.** The spectral WCS (`WCS-WAVE`, key `W`) gives the wavelength at the target's pixel, reported with every frame.

2.5 One grid for every frame

The grid is a gnomonic (TAN) projection, north up and east left, with SPHEREx-sized pixels. It is centred on the target's position at the mean date of all exposures and has an odd number of pixels per side, $N = \mathrm{round}(\mathrm{fov}/p)$ rounded up to odd, so that the centre falls on a pixel. Because the grid is fixed on the sky and not on the target, the field stars stay still and a moving target walks across the grid. For every grid pixel the sky position is converted to the pixel coordinates of the exposure through the exposure's own WCS, and the image is sampled there with bilinear interpolation. The variance is sampled with nearest neighbour, so it is not smoothed.

2.6 Passes

SPHEREx covers a given point in visits a few days long, about six months apart. The frames are split into passes wherever two consecutive exposures are more than 60 days apart. The pass index is what the viewer uses for its default mode (Section 4).

3 The server

`tool/server.py` uses the Python standard library only, besides the packages the data path already needs (`numpy`, `scipy`, `astropy`). It serves the page in `tool/web/` and one endpoint:

```
GET /api/frames?ra=..&dec=..&size=..&band=..&epoch=..&pmra=..&pmdec=..&name=..&mask=..
```

Inputs are checked ($0 \le \alpha < 360$, $-90 \le \delta \le 90$, field side between 20 and 600 arcsec, detector D1 to D6) and a bad input returns HTTP 400 with the reason. An archive failure returns HTTP 502 with the error text, so the page can tell the user what went wrong. Results are kept in memory per view, and when the same view is asked twice at once the second request waits for the first download instead of starting its own.

4 The page

The page (`tool/web/index.html` and `app.js`) draws the frames on a canvas, pixel for pixel, with no smoothing.

Two modes. In `pass` mode, the default, each pass is reduced to its per-pixel median, and the blink steps through the passes. This removes most cosmic rays and single-exposure artefacts, and it matches the six-month rhythm in which a proper motion becomes visible. In `frames` mode every exposure is shown on its own, with its date, pass, wavelength and file name.

Stretch. Each image is stretched on its own, so the change of brightness with wavelength between exposures does not swamp the blink. The sky level m is the median of the finite pixels and the noise is the scaled median absolute deviation, $\sigma = 1.4826\,\mathrm{MAD}$. In the default `asinh` stretch a pixel value v becomes

$$t = \frac{\operatorname{arsinh}\left((v-\ell)/\beta\right)}{\operatorname{arsinh}\left((h-\ell)/\beta\right)}, \qquad \ell = m - 2\sigma, \quad \beta = 3\sigma, \tag{3}$$

clipped to $[0, 1]$, where h is the upper percentile set by the `contrast` parameter. This keeps faint field stars visible next to a saturated bright one. A linear stretch between two percentiles is also available.

Marker and compass. A red marker of four short ticks, with a gap in the middle so it never covers the object, shows where the target should be at each date. N and E labels show the orientation.

The URL is the state. As in WiseView, every setting is a URL parameter, and the URL is rewritten when a control changes. Table 1 lists them. Presets fill in the position, motion and detector for the five challenge targets, for example `/?target=Barnard`.

Parameter	Meaning	Default
`target`	preset name: `Barnard`, `EpsIndB`, `WISE0855`, `RWAur`, `RDra`	
`ra`, `dec`	position in degrees at `epoch`	
`pmra`, `pmdec`	proper motion in mas yr^{-1}	0
`epoch`	epoch of the position, decimal year	2016.0
`size`	field side in arcsec	108
`band`	SPHEREx detector, D1 to D6	D1
`mode`	`pass` (median of each pass) or `frames` (every exposure)	`pass`
`speed`	blink interval in ms	500
`stretch`	`asinh` or `linear`	`asinh`
`contrast`	upper percentile of the stretch	99.5
`invert`, `marker`, `mask`	1 or 0	0, 1, 0

Table 1: URL parameters. Keyboard: space plays or pauses, the arrow keys step.

5 Running it

```
git clone https://github.com/yourastrophysicist/SPHEREx-site
cd SPHEREx-site/tool
conda activate astrolab2 # any Python 3.10+ with numpy, scipy, astropy
python server.py
# open http://localhost:8000/?target=Barnard
```

The first view of a new field asks IRSA for every exposure and downloads the cutouts. This takes a few minutes, and up to ten when the archive is busy. The files are then kept in `~/spherex_data` (or the folder in `$SPHEREX_DATA`) and the same view opens in seconds. The port can be changed with `$PORT`.

6 Tests on the challenge targets

Table 2 gives the inputs of the five presets. Positions and motions are from Gaia DR3 at J2016.0, except WISE 0855−0714, which is too faint for Gaia and uses the AllWISE position with the motion of Kirkpatrick et al. (2021). The detector is the one where the target is easiest to see.

Target	α (deg)	δ (deg)	epoch	$\mu_{\alpha*}$ (mas yr^{-1})	μ_δ	detector
Barnard's Star	269.448503	4.739420	2016.0	−801.6	10362.4	D1
Epsilon Indi B	331.076453	−56.793812	2016.0	3982.0	−2466.8	D5
WISE 0855−0714	133.794760	−7.245146	2010.56	−8123.7	673.2	D6
RW Aurigae	76.956081	30.401205	2016.0	2.6	−27.4	D1
R Draconis	248.167378	66.755007	2016.0	−20.6	6.1	D3

Table 2: The five presets.

Barnard's Star, D1. 60 exposures over 3 passes, on a 19×19 grid ($108''$ field). The measured position of the star sits 0.06 px from the position predicted by the proper motion, which checks the alignment and the motion correction together. The core of the star is saturated and flagged (57 grid pixels in the masked view), which is why the viewer draws unmasked pixels by default. In the blink the star walks north by about $10.4''$ per year, which is 0.85 px per six-month pass, and the walk is plain to see over the three passes.

WISE 0855−0714, D6. The coldest known brown dwarf is faint at SPHEREx's longest wavelengths but visible in the pass medians. The preset opens a wider field, 216″, to give more field stars to judge against.

RW Aurigae and R Draconis. These two barely move. They are in the set to show the other half of the job: a young variable star and a Mira variable whose brightness changes between passes while the field around them stays the same.

7 What it can and cannot find

Published forecasts put a single SPHEREx exposure at roughly $V \approx 17$ for a point source at 5σ per channel, and the all-sky map after four passes at about AB 19.5 to 19.9 (deeper near the ecliptic poles). These are forecasts, not our measurements. The blink works on single passes, so its practical limit is close to the single-exposure depth.

Behind the challenge sits the question of a distant "Planet X". For a planet seen in reflected sunlight at opposition the flux falls as r^{-4}:

$$m = m_\odot - 2.5 \log_{10} \frac{A_g R_p^2}{r^2 \Delta^2}, \qquad m_\odot = -26.74 \ (V), \tag{4}$$

with the radius R_p and the distances r and $\Delta \approx r$ in AU and A_g the geometric albedo. A planet of three Earth radii with $A_g = 0.3$ is at $V \approx 18.8$ at 300 AU and $V \approx 21.0$ at 500 AU. That is fainter than a single SPHEREx exposure, so this tool will not find a Planet Nine of the usual size and distance, and we say so on the page. What it does well is the thing SPHEREx is suited for: nearby stars and brown dwarfs with large proper motions, and variable stars, at infrared wavelengths and on the whole sky.

Known limits of v1.0.0.

- Linear proper motion only, no parallax.
- One detector at a time. The wavelength changes slightly between exposures because of the linear variable filter. It is reported, not corrected.
- The first view of a new field is slow because the cutouts come from IRSA live.
- The server is meant to run locally or behind a single host. It has no rate limiting.

8 Use of earlier work and of AI

Earlier work. The data path grows out of two prep scripts and a notebook made by Jessica Syafaq Muthmaina on 10 November 2026, before the event, to check that the SPHEREx data could be reached. The TAP query, the proper-motion centring, the retrying download and the flag mask come from there. The Milano Local Lead confirmed on 14 November 2026 that this earlier work may be used. It is kept apart in the `prep/` folder of the repository, with its own README and guide.

AI. Parts of the code and of this document were written with help from Claude (Anthropic). The team chose the method, checked the results against the archive and the expected motions, and is responsible for the content.

9 Credits and data use

- **SPHEREx** Level 2 spectral images: NASA/JPL-Caltech/IPAC, through the NASA/IPAC Infrared Science Archive (IRSA).
- **Gaia** DR3: European Space Agency, mission Gaia, processed by the Gaia Data Processing and Analysis Consortium (DPAC).
- **AllWISE**: NASA/JPL-Caltech, UCLA. Motion of WISE 0855−0714 from Kirkpatrick et al. (2021).
- **WiseView** (Caselden et al.), used as the model for the interface.
- Software: NumPy, SciPy, Astropy.

NASA does not endorse this project. Data are used under the terms of each archive.

References

[1] Bock, J. et al., SPHEREx mission, NASA/JPL-Caltech/IPAC, `https://spherex.caltech.edu`.

[2] NASA/IPAC Infrared Science Archive, SPHEREx data and tutorials, `https://irsa.ipac.caltech.edu` and `https://caltech-ipac.github.io/irsa-tutorials`.

[3] Gaia Collaboration, Vallenari, A. et al. 2023, *Gaia Data Release 3: Summary of the content and survey properties*, A&A 674, A1.

[4] Kirkpatrick, J. D. et al. 2021, *The field substellar mass function based on the full-sky 20 pc census of 525 L, T, and Y dwarfs*, ApJS 253, 7.

[5] Caselden, D. et al. 2018, *WiseView: Visualizing motion and variability of faint WISE sources*, Astrophysics Source Code Library, ascl:1806.004.

[6] Astropy Collaboration 2022, *The Astropy Project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0)*, ApJ 935, 167.

[7] Earlier work: `prep/` in `https://github.com/yourastrophysicist/SPHEREx-site`, 10 November 2026.